



MesoNET cluster

McStas and McXtrace on juliet

Hardware layout and tips

McStas — neutrons

McXtrace — X-rays



Peter Willendrup · DTU Physics

PaNRAID School · supported by PEPR DIADEM

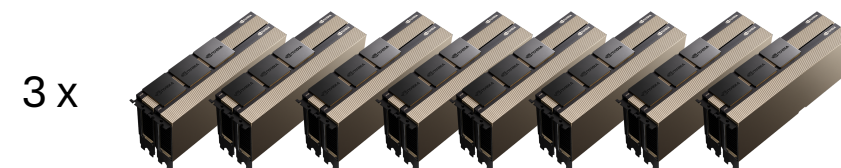
NVIDIA-SMI 510.43.02					KMD Version: 610.43.02		CUDA UMD Version: 13.3	
GPU Fan	Name Temp	Perf	Persistence-M Pwr:Usage/Cap	Bus-Id	Disp.A Memory-Usage	Volatile GPU-Util	Uncorr. Compute M. MIG M.	ECC M.
0 N/A	NVIDIA 35C	A100-SXM4-80GB P0	65W / 400W	00000000:07:00.0	Off 0MiB / 81920MiB	0%	0 Default Disabled	
1 N/A	NVIDIA 34C	A100-SXM4-80GB P0	64W / 400W	00000000:0B:00.0	Off 0MiB / 81920MiB	0%	0 Default Disabled	
2 N/A	NVIDIA 32C	A100-SXM4-80GB P0	63W / 400W	00000000:48:00.0	Off 0MiB / 81920MiB	0%	0 Default Disabled	
3 N/A	NVIDIA 35C	A100-SXM4-80GB P0	62W / 400W	00000000:4C:00.0	Off 0MiB / 81920MiB	0%	0 Default Disabled	
4 N/A	NVIDIA 32C	A100-SXM4-80GB P0	60W / 400W	00000000:88:00.0	Off 0MiB / 81920MiB	0%	0 Default Disabled	
5 N/A	NVIDIA 35C	A100-SXM4-80GB P0	63W / 400W	00000000:8B:00.0	Off 0MiB / 81920MiB	0%	0 Default Disabled	
6 N/A	NVIDIA 34C	A100-SXM4-80GB P0	61W / 400W	00000000:C8:00.0	Off 0MiB / 81920MiB	0%	0 Default Disabled	
7 N/A	NVIDIA 34C	A100-SXM4-80GB P0	62W / 400W	00000000:CB:00.0	Off 0MiB / 81920MiB	0%	0 Default Disabled	

Processes:						
GPU	GI ID	CI ID	PID	Type	Process name	GPU Memory Usage
No running processes found						



3 x NVIDIA GPU boxes
(juliet2,3,4)

each with 8 x A100 cards
- 24 GPU's to share

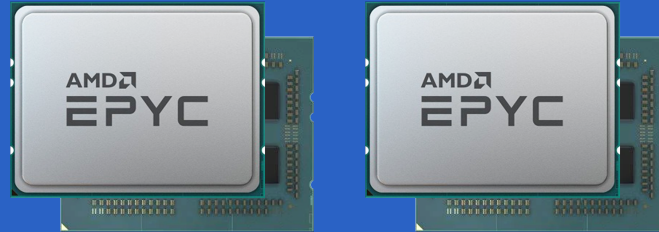


Reservation for PaNRAID:
m26216

1000 GPU hours
(5 days x 8 hours x 24 GPU)

Our reservation

```
processor      : 111
vendor_id     : AuthenticAMD
cpu family    : 25
model         : 1
model name    : AMD EPYC 7663 56-Core Processor
stepping      : 1
microcode     : 0xa0011de
cpu MHz       : 1996.251
cache size    : 512 KB
physical id   : 1
siblings      : 56
core id       : 62
cpu cores     : 56
apicid        : 126
initial apicid : 126
fpu           : yes
fpu_exception : yes
cpuid level   : 16
wp            : yes
flags         : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse3
pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc cpuid extd_apicid
sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand lahf_lm cmp_legacy extapic cr8_
wdt tce topoext perfctr_core perfctr_nb bpeext perfctr_llc mwaitx cpb cat_l3 cdp_l3 hw
avx2 smep bmi2 erms invpcid cqm rdt_a rdseed adx smap clflushopt clwb sha_ni xsaveopt
cqm_mbm_local user_shstk clzero irperf xsaveerptr rdpru wbnoinvd amd_ppin brs arat np
decodeassists pausefilter pfthreshold v_vmsave_vmload vgif v_spec_ctrl umip pku ospke
debug_swap
bugs          : sysret_ss_attrs spectre_v1 spectre_v2 spec_store_bypass srsr ibpb_no_ret s
bogomips      : 3992.50
TLB size      : 2560 4K pages
clflush size   : 64
cache_alignm   : 64
address sizes  : 48 bits physical, 48 bits virtual
power managem : ts ttp tm hwpstate cpb eff_freq_ro [13] [14]
```



3 nodes x 2 x 56 cores pr node

The nodes each have 112 CPU cores

- only 1 core is active pr
active GPU if running
McStas / McXtrace
OpenACC

- also share ;-)

Reservation for PaNRAID:
m26216

When to choose CPU x MPI over GPU?

Issues when for example...

- * McStas or McXtrace instrument includes **Union** components
- * McStas instruments is using **NCrystal_sample**
- * McXtrace components require **xraylib**

... the code-generator will tell you if you are in **FUNNEL** mode...

Our reservation

```
mcrun -c --openacc PSI_DMC.instr -n0
INFO: No output directory specified (--dir)
INFO: Using directory: "PSI_DMC_20260919_185436"
INFO: Regenerating c-file: PSI_DMC.c
Info: 'Al_window' is a contributed component.
Info: 'Monochromator_2foc' is a contributed component.
```

WARNING:

The parameter format of sample is initialized using a static {,,,} vector.

- > Such static vectors support literal numbers ONLY.
- > Any vector use of variables or defines must happen via a DECLARE/INITIALIZE pointer.

Generating single GPU kernel or single CPU section layout:

- > SPLIT N at component foc_mono
- > SPLIT N at component sample

Generating GPU/CPU -DFUNNEL layout:

- > SPLIT within buffer at component foc_mono
- > SPLIT within buffer at component sample

CFLAGS= @NCRYSTALFLAGS@



* Not seeing -DFUNNEL
in the final cogen CFLAGS
line basically means:

Full GPU acceleration



```
mxrun -c --openacc SOLEIL_ROCK.instr -n0  
INFO: No output directory specified (--dir)  
INFO: Using directory: "SOLEIL_ROCK_20260919_184807"  
INFO: Regenerating c-file: SOLEIL_ROCK.c  
Info: 'Mirror_toroid_pothole' is a contributed component.
```

Generating single GPU kernel or single CPU section layout:
-> SPLIT N at component absorption_sample

Generating GPU/CPU -DFUNNEL layout:
Component absorption_sample is NOACC, CPUONLY=1
-> FUNNEL mode enabled, SPLIT within buffer.
-> SPLIT within buffer at component absorption_sample
-> GPU kernel from component fluo_monitor

CFLAGS= -std=c99 @XRLFLAGS@ -DUSE_OFF -DFUNNEL



* -DFUNNEL basically means:

Better use CPU's

